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                        LED BLINK BUILD SAMPLE
                        Nucleus STM32 Platform
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STEP 1: PROJECT INITIALIZATION
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```
$ nucleus init --board NUCLEO-F411RE
created  stm32.toml
created  CMakeLists.txt
created  cmake/arm-none-eabi-gcc.cmake
created  STM32F411RETx_FLASH.ld
created  src/main.c
created  src/stm32f4xx_hal_conf.h
created  src/stm32f4xx_it.h
created  src/stm32f4xx_it.c
created  .github/workflows/ci.yml
created  .gitignore
```

Scaffolded 10 file(s) for NUCLEO-F411RE.

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STEP 2: CONFIGURE LED ON PA5
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File: stm32.toml

```
=====
[device]
family   = "STM32F411RE"
board    = "NUCLEO-F411RE"
clock_hz = 100_000_000
```

```
[build]
toolchain = "arm-none-eabi-gcc"
optimization = "Os"
```

```
[peripherals.usart2]
tx  = "PA2"
rx  = "PA3"
baud = 115200
=====
```

File: src/main.c (LED Blink Code)

```
=====
#include "stm32f4xx_hal.h"
#include "generated/nucleus_config.h"

int main(void)
{
    HAL_Init();
    SystemClock_Config();
    Nucleus_Init();

    __HAL_RCC_GPIOA_CLK_ENABLE();

    GPIO_InitTypeDef led = {0};
    led.Pin   = GPIO_PIN_5;
    led.Mode  = GPIO_MODE_OUTPUT_PP;
    led.Pull  = GPIO_NOPULL;
    led.Speed = GPIO_SPEED_FREQ_LOW;
    HAL_GPIO_Init(GPIOA, &led);

    while (1) {
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5); // Toggle LD2 (green LED)
        HAL_Delay(500);                          // 500ms blink period
    }
}
=====
```

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STEP 3: VALIDATE CONFIGURATION
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```
$ nucleus check
stm32.toml OK - 1 peripheral(s), no conflicts.
```

✓ Configuration valid

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=====
STEP 4: BUILD FIRMWARE
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```
$ nucleus build
generated ./src/generated/nucleus_config.h
generated ./src/generated/nucleus_init.c
-- The C compiler identification is GNU 14.2.0
-- The ASM compiler identification is GNU
-- Found assembler: /usr/bin/arm-none-eabi-gcc
-- Configuring done (0.4s)
-- Generating done (0.0s)
-- Build files have been written to: ./build
[ 4%] Building C object CMakeFiles/firmware.dir/src/main.c.obj
[ 8%] Building C object CMakeFiles/firmware.dir/src/stm32f4xx_it.c.obj
[12%] Building C object CMakeFiles/firmware.dir/src/generated/nucleus_init.c.obj
[16%] Building ASM object CMakeFiles/firmware.dir/.../startup_stm32f411xe.s.obj
[20%] Building C object CMakeFiles/firmware.dir/.../system_stm32f4xx.c.obj
[25%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal.c.obj
[29%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_rcc.c.obj
[33%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_rcc_ex.c.obj
[37%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_gpio.c.obj
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```
[ 41%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_exti.c.obj
[ 45%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_cortex.c.obj
[ 50%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_pwr.c.obj
[ 54%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_pwr_ex.c.obj
[ 58%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_flash.c.obj
[ 62%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_flash_ex.c.obj
[ 66%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_dma.c.obj
[ 70%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_dma_ex.c.obj
[ 75%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_uart.c.obj
[ 79%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_spi.c.obj
[ 83%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_i2c.c.obj
[ 87%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_i2c_ex.c.obj
[ 91%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_tim.c.obj
[ 95%] Building C object CMakeFiles/firmware.dir/.../stm32f4xx_hal_tim_ex.c.obj
[100%] Linking C executable firmware
Generating firmware.bin / firmware.hex
[100%] Built target firmware
```

```
✓ Build successful!
Output: ./build/firmware.elf
        ./build/firmware.bin
        ./build/firmware.hex
```

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RESULT: READY TO FLASH
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```
$ ls -lh build/firmware.*
-rw-r--r-- 1 harsh harsh 47K Jun 23 19:15 ./build/firmware.bin
-rw-r--r-- 1 harsh harsh 52K Jun 23 19:15 ./build/firmware.elf
-rw-r--r-- 1 harsh harsh 62K Jun 23 19:15 ./build/firmware.hex
```

```
✓ Firmware compiled and ready for flashing to NUCLE0-F411RE
✓ LED on PA5 will blink at 500ms intervals

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```